

Line Drawing Algorithms (DDA)

Need for Line Drawing Algorithm

In raster graphics, a straight line cannot be drawn directly. Line drawing algorithms determine which pixel positions should be selected to represent a straight line.

Types of Line Drawing Algorithms

1. DDA Line Drawing Algorithm
2. Midpoint Line Drawing Algorithm
3. Bresenham Line Drawing Algorithm

DDA (Digital Differential Analyzer) Algorithm

DDA is an incremental scan conversion algorithm that generates points of a line at unit intervals.

Given

Starting point: (x_1, y_1)

Ending point: (x_2, y_2)

Steps of DDA Algorithm

Step 1:

$$\Delta x = x_2 - x_1$$

$$\Delta y = y_2 - y_1$$

Step 2:

If $|\Delta x| > |\Delta y|$

$$\text{steps} = |\Delta x|$$

Else

$$\text{steps} = |\Delta y|$$

Step 3:

Let current point be (x_p, y_p)

$$\text{Slope } m = \Delta y / \Delta x$$

Case 1: $m < 1$ $x_{p+1} = \text{round}(x_p + 1)$ $y_{p+1} = \text{round}(y_p + m)$	Case 2: $m = 1$ $x_{p+1} = \text{round}(x_p + 1)$ $y_{p+1} = \text{round}(y_p + 1)$	Case 3: $m > 1$ $x_{p+1} = \text{round}(x_p + 1/m)$ $y_{p+1} = \text{round}(y_p + 1)$
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Step 4:

Repeat Step 3 until end point is reached.

Example

Starting point: (1,7)

Ending point: (11,17)

Solution (Using DDA Line Drawing Algorithm)

Step 1: Calculate Differences

$$\Delta x = x_2 - x_1 = 11 - 1 = 10$$

$$\Delta y = y_2 - y_1 = 17 - 7 = 10$$

Step 2: Determine Number of Steps

$$\text{steps} = \max(|\Delta x|, |\Delta y|)$$

$$\text{steps} = \max(10, 10) = 10$$

Step 3: Calculate Increment Values (Case 3 $m=1$)

$$x_{p+1} = \text{round}(x_p + 1)$$

$$y_{p+1} = \text{round}(y_p + 1)$$

Step 4: Initial Point

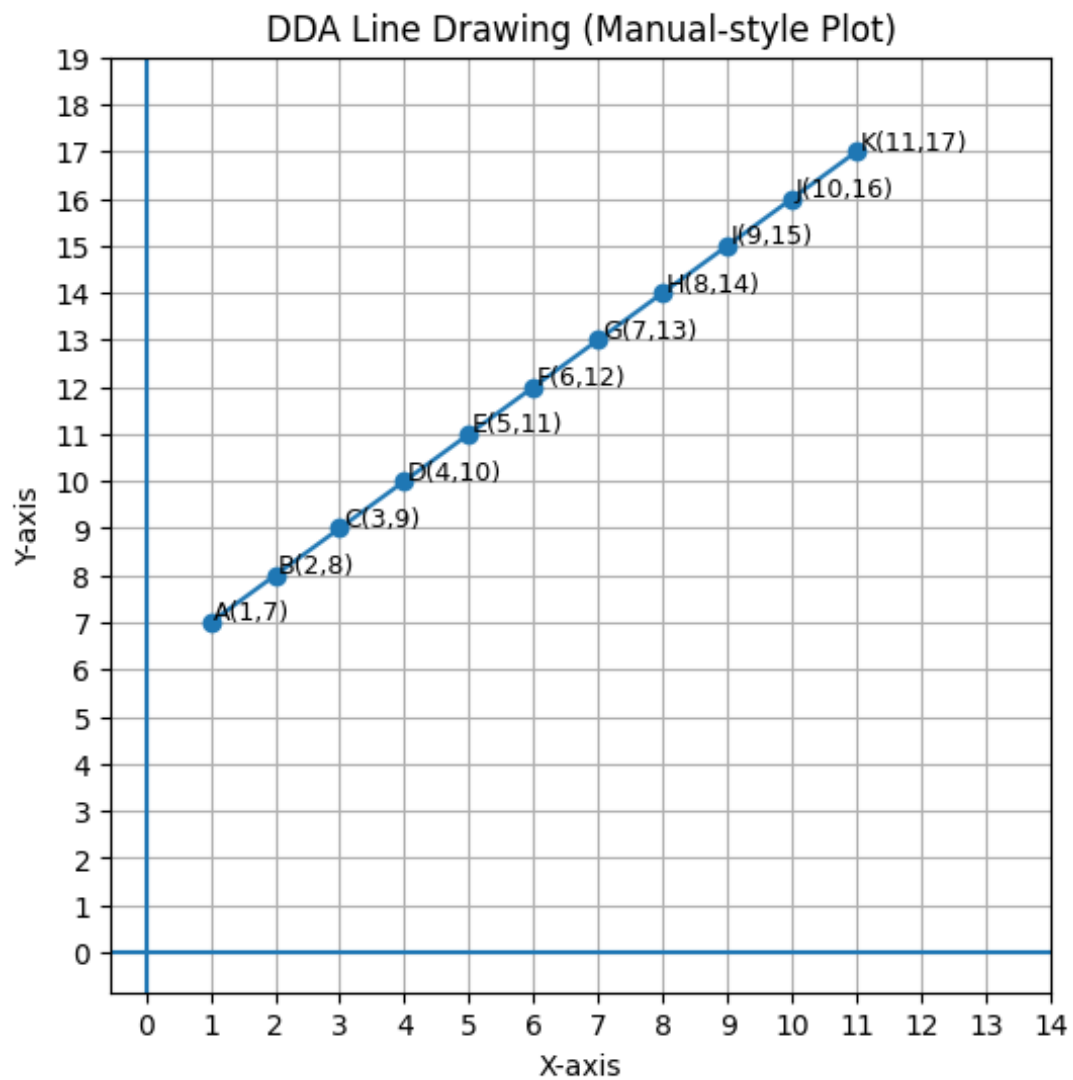
$$x = 1, y = 7$$

Generated Points

Step	x	y	Plot Points
Initial -	1	7	(1,7)
1	2	8	(2,8)
2	3	9	(3,9)
3	4	10	(4,10)
4	5	11	(5,11)
5	6	12	(6,12)
6	7	13	(7,13)
7	8	14	(8,14)
8	9	15	(9,15)
9	10	16	(10,16)
10	11	17	(11,17)

Generated points:

(1,7), (2,8), (3,9), (4,10), (5,11), (6,12), (7,13), (8,14), (9,15), (10,16), (11,17)



Advantages

- Simple and easy
- Works for all slopes
- Easy to implement

Disadvantages

- Uses floating-point operations
- Slower than Bresenham algorithm

Homework: DDA Line Drawing Algorithm

Problem 1

Using the **DDA line drawing algorithm**, draw a line between the following two points and generate all intermediate points:

$(x_1, y_1) = (2, 3)$

$(x_2, y_2) = (6, 15)$

Problem 2

Using the **DDA line drawing algorithm**, draw a line between the following two points and generate all intermediate points:

$(x_1, y_1) = (1, 4)$

$(x_2, y_2) = (11, 8)$